

# AHMED BARGOUGUI

AI & Data Engineering | Backend Developer | Computer Science — Final Year  
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## PROFILE

**Final-year Computer Science student** with two internships at Sagemcom — **Telecom AI Engineering (RAG, OCR, CrewAI, n8n automation, validation pipelines)** and Business Intelligence. Built production backend systems used by **32,000+ users** and AI-powered applications. Strong interest in **end-to-end AI pipelines, generative AI, and scalable backend architecture**. Seeking a PFE internship in AI or Backend Engineering.

## PROFESSIONAL EXPERIENCE

**Telecom Data & AI Engineering Intern** — Sagemcom • Ben Arous, Tunisia 07/2026 – 08/2026

- Analysed and structured telecom product data through product discovery, inventory creation, **OCR-based extraction**, and technical data processing
- Contributed to an **ICP/MTP validation** application comparing technical specifications against test data — generating structured PASS/FAIL compliance reports with conformities, missing/extra items, and limit mismatches
- Integrated **CrewAI** as a Generative AI analysis layer to interpret validation results and generate automated conclusions — designed fail-safe to not block validation on AI errors
- Built an automated **n8n + Gmail** reporting pipeline connecting application outputs to production webhooks and delivering final validation reports by email
- Designed an end-to-end workflow: data processing → OCR → validation → Generative AI → n8n automation → automated email reporting

**Business Intelligence Engineer Intern** — Sagemcom • Ben Arous, Tunisia 07/2025 – 08/2025

- Designed **Star Schema** data warehouse + ETL pipelines (Pandas, SQLAlchemy) automating data ingestion from multiple sources
- Delivered **Power BI** dashboards for real-time KPI tracking and standardised 10,000+ records via Power Query

## PROJECTS

**ScanVD — AI Video Search Engine** — huggingface.co/spaces/AhmedBrg/ScanVD\_demo 2026

- Multi-model AI pipeline (**Whisper + YOLOv8 + CLIP**) extracting precise timestamps from natural language queries — deployed on HuggingFace with Docker

**Stack:** Python • Whisper • YOLOv8 • CLIP • HuggingFace • Docker

**Secure Auth System — Gaming Community** — Freelance • Private 2026

- Architected and deployed a **secure REST API** (Node.js, Express, MySQL) for a 32,000-member gaming community — sole engineer, full lifecycle ownership
- Built a C# desktop launcher with HWID binding, Discord OAuth2, JWT session management, rate limiting, and API validation

**Stack:** Node.js • Express • MySQL • C# • JWT • OAuth2 • Git

**AgriLink — Smart Farming Platform** — TGN Hackathon 2025 • github.com/Ahmedbrg/Hackathon2025 2025

- Built a mobile app for **AI-based plant disease detection** with GPS networking and offline-first architecture for rural environments

**Stack:** React Native • TypeScript • AI API • GPS

## TECHNICAL SKILLS

|                    |                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------|
| <b>AI / GenAI:</b> | RAG • LLMs • CrewAI • Qdrant • Weaviate • Whisper • YOLOv8 • CLIP • OCR • Embeddings • Ollama |
| <b>Automation:</b> | n8n • Webhooks • REST APIs • Gmail API • Workflow Automation • JSON pipelines                 |
| <b>Backend:</b>    | Node.js • Express • FastAPI • Auth Systems • JWT • OAuth2 • MySQL • C#                        |
| <b>Data / BI:</b>  | SQL Server • ETL • Pandas • SQLAlchemy • Power BI • Power Query • Star Schema                 |
| <b>Languages:</b>  | Python • JavaScript • Java • C++ • C# • SQL                                                   |
| <b>Tools:</b>      | Git • Docker • HuggingFace • Linux • Postman • VS Code • Jupyter                              |

## EDUCATION • LEADERSHIP • LANGUAGES

**Bachelor's Degree in Computer Science** — FSTSBZ, University of Kairouan 09/2024 – 06/2027

Final Year (PFE) • Algorithms • Data Structures • Databases • OOP • OS • Networks • AI

- Founder & President, Club DevUp** — AI/ML & career workshops, 50+ students (11/2025–05/2026) • **Trainer & Jr. Secretary General, JCI Sidi Bouzid** (2023–Present)
- Ranked 3rd — ACM-style Competitive Programming Contest** • **Arabic** Native • **French** Professional • **English** Professional